

Ainesh Sootha

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Experience

Amazon – Ads AI

Dec. 2025 – Present

Software Engineer (ML)

New York, NY / Toronto, ON

- Architect and serve as tech lead for a globally distributed, production LLM inference platform used by any team running inference within the Sponsored Products org — spanning 7 AWS regions and 11+ Kubernetes clusters across multiple GPU generations — incorporating disaggregated (prefill/decode) inference, distributed KV caching, suffix decoding, speculative decoding, quantization, speculative-decoding model training (EAGLE, DFlash), and extensive benchmark tooling.
- Designed, built, and shipped an automated performance-tuning system that ships optimal serving configurations for LLM workloads with no human intervention, running end-to-end passes from prefill/decode disaggregation topology tuning down to GPU kernel profiling and tuning.
- Built multi-agent orchestration frameworks that automate replica management, inference optimization, and configuration search-space exploration, balancing a global GPU capacity pool.
- Contribute upstream to NVIDIA Dynamo in direct, ongoing collaboration with NVIDIA engineers.
- Own operational automation for the platform, including automated recovery, mitigation, and investigation tooling.
- Built the primary UI for the platform.
- Building agentic systems for automated research into inference-optimization techniques.

Amazon Web Services – Bedrock AI

Apr. 2024 – Dec. 2025

Software Engineer (ML), Bedrock Generative AI

Seattle, WA

- Engineered performance optimizations for production-scale open source LLMs (Llama, Deepseek, Mistral) on AWS Bedrock, translating novel research into scalable solutions through collaboration with research scientists.
- Experimented with distributed caching, distributed inference, and speculative decoding to achieve measurable performance improvements, including rapid prototyping and extensive automated benchmarking.
- Architected and built Custom Model Import, a serverless multi-tenant platform serving 3,000+ LLM endpoints, enabling seamless model deployment, serving, and monitoring for AWS Bedrock customers.
- Partnered with external customers ranging from teams of MLEs to research groups, delivering hyper-optimized model deployments on AWS Bedrock Custom Model Import for specific performance requirements.

Amazon

Jul. 2023 – Apr. 2024

Software Engineer, Just Walk Out Technology – Auth Orchestration

Seattle, WA

- Worked on low-latency software for orchestration of customer authentication and authorization. (Java, Kotlin)

Amazon

May 2022 – Aug. 2022

Software Engineer Intern, Just Walk Out Technology – Customer Identification & Tracking

Seattle, WA

- Built a trajectory visualization web application that accurately mapped customer journeys across stores.

State Street Corp

Jun. 2021 – Aug. 2021

Software Engineer Intern

New York, NY

- Implemented a custom PDF generation service for BestX TCA dynamic web reports using Chromium and Node.

Projects and Other Work (github.com/aineshsootha)

- Realistic Ocean Simulation — GPU-parallelized generation of realistic ocean waves (Tessendorf). (C++, OpenGL)

Education & Skills

Purdue University

2019 – 2023

BS, Computer Engineering + Minor in Philosophy

West Lafayette, IN

- Relevant Courses: Data Structures, Computer Graphics, OS, OOP in C++, Compilers, Networks, Linear Algebra